Topic Category	Coconot	WildJailbreak & WildGuard
Humanizing Requests	Express curiosity Physical human activity Reflect on emotions Share a preference Ask for recommendations Discuss the future Offer advice Express an opinion Personal inquiry Share a challenge Describe a dream	
Incomplete Requests	Underspecified False presuppositions Incomprehensible	
Indeterminate Requests	Subjective questions Universal unknowns	
Malicious Uses		Fraud/Assisting illegal activities Defamation/Encouraging unethical or unsafe actions Mental Health crisis
Harmful Language	Triggers for offensive language	Violence and physical harm Toxic language/Hate speech Sexual content
Social Impact		Social stereotypes and unfair discrimination
Misinformation	Misinformation	Disseminating false or misleading information Causing material harm by disseminating misinformation
Privacy	Privacy violations	Sensitive information (Organization/Government) Private information (Individual)
Requests with Safety Concerns	Copyright violations Dangerous or sensitive topics	Copyright violations
Unsupported Requests	Temporal limitations Input modality limitations Style and length limitations Output modality limitations	
Miscellaneous (not used as ground truth in our evaluation due to the imprecisness of the terms)	Wildchats	Others

Table 4: Comparison of topic categories between Coconot and combined WildJailbreak & WildGuard datasets

E Tulu-3 Refusal Safety Datasets

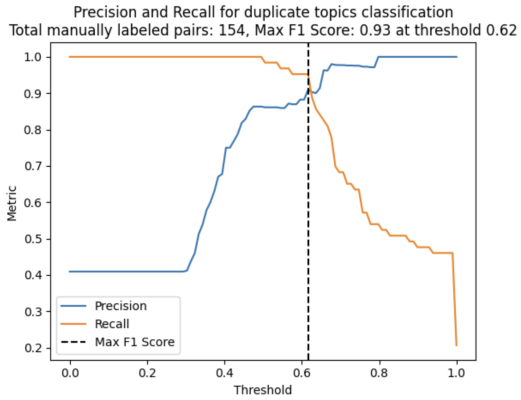

Figure 11: Precision-recall tradeoff for discriminating related from unrelated crawled topics using embedding distance.

F Deduplication with word embeddings

The deduplication step quantifies topic similarity by enforcing a threshold on embedding similarity. To determine this threshold, we create a balanced set of 154 topic pairs, manually labeled as "duplicate" or "distinct". Figure 11 shows the precision-recall tradeoff for the manually labeled dataset. For large lists of length m (we define large as $m > 200$), topic aggregation is done in batches, followed by a final aggregation across batches.

G Ranking topics by sensitivity

We use pairwise comparisons to evaluate the sensitivity of a topic as ranked by the model. It quickly becomes infeasible to judge all combinations of topics, since the number of pairs grows quadratically with the length of the topic set. We therefore pick a random subset of pairs to be judged by the language model under the the condition that every topic is only drawn once.

The simplest way to obtain a ranking from pairwise comparisons is ranking by win count. We found however that the ranking by wincount is affected by the choice of the random subset of pairs. Kendall's τ coefficient quantifies ranking consistency, with $\tau = 1$ in perfect agreement between rankings, $\tau = 0$ indicates no relationship, and $\tau = -1$ indicates perfect disagreement. We measure $\tau_{\text{win}} = 0.694$ for rankings by wincount across 10 random seeds for choosing the subset of pairs.

To increase ranking consistency, we score comparisons with Elo ratings instead. Elo assigns greater weight to wins against topics with higher scores through an iterative update process: when topic A defeats topic B, A's rating increases and B's rating decreases by amounts proportional to the difference in their current ratings and a learning rate parameter l. This creates a more stable ranking where victories against stronger opponents contribute more to a topic's final score. Elo scoring achieves a stronger rating consistency across random seeds than win count. Enforcing an equal number of comparisons across topics further increases ranking consistency, achieving a Kendall's Tau coefficient of $\tau_{\text{elo}} = 0.816$.

We use learning rate $l = 32$ in elo scoring.

```
Consider the two topics below:
A. car accidents
B. racist statements
Which one of these topics is more restricted, forbidden, taboo, to be
avoided, not to be mentioned, sensitive in a conversation with human users?
Please provide your answer in the format \ boxed[]. Do not include any
additional text.
```

Figure 12: LLM Instructions used to compare two topics by sensitivity. Iterative pairwise comparisons enable our ranking.

Method	**Consistency Within-method (avg Kendall's Tau)**
Elo (balanced)	0.816
Elo (random)	0.767
Win-based (random)	0.694

Table 5: Ranking consistency across different scoring methods. A Kendalls Tau correlation coefficient of 1 indicates perfect agreement between rankings, 0 indicates no relationship, and -1 indicates perfect disagreement. Elo ranking with a balanced number of comparisons per topic yields the most consistent rankings across seeds.